

## RESEARCH ARTICLE OPEN ACCESS

# Simulation-Based Analysis of Silicon Solar Cell Performance with Laser Enhanced Contacts: Impact of Bulk Defect Density and Resistivity

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## ABSTRACT

Laser-enhanced metal contact technology has recently emerged as an effective approach to reducing contact recombination in silicon solar cells, particularly in tunneling oxide passivating contacts cells with a front boron-doped emitter and rear phosphorus-doped polysilicon based passivating contact. This technique enables superior front-side surface passivation and open-circuit voltages comparable to those of silicon heterojunction counterparts. In this study, we conducted a comprehensive simulation-based analysis comparing the devices with laser-enhanced contacts (LASER) to conventional devices with selective emitters, using experimentally extracted bulk defect parameters across a range of bulk resistivities. Additionally, we evaluated the low-light illumination response of the devices and conducted energy yield simulations under various solar conditions. High-resistivity wafers consistently enhance efficiency when bulk defect levels are low but may degrade performance when defect densities are high, especially in devices with laser-enhanced contacts. Under low-light conditions, the benefits of high-resistivity wafers are further diminished in the presence of significant bulk defects, resulting in reduced energy yields in regions with poor or variable solar resources, despite gains in areas with abundant sunlight. The findings provide valuable insights into the impacts of bulk resistivity, bulk defect density, and illumination intensity on the device performance as well as energy yield.

## 1 | Introduction

In recent years, laser-enhanced metal contact technology has emerged as a promising approach to reducing metal/silicon contact recombination in silicon (Si) solar cells [1–7], which was early applied to passivating emitter rear contact (PERC) solar cells and then employed to tunneling oxide passivating contacts (TOPCon) solar cells as a crucial strategy. This technology involves applying highly intense local laser illumination near the metal contacts close to the p–n junction of the device under reverse bias conditions [1, 2]. By utilizing laser-enhanced contacts (LASER), the need for selective emitters (SE) to ensure good

contacts between metal and silicon is eliminated, thereby simplifying the fabrication process. The laser-enhanced process creates a large quantity of microscale contacts [6], forming excellent metal/silicon contacts and significantly mitigating the overall metal/silicon contact recombination by reducing the fraction of direct-contact interface area. Moreover, this technology allows for the use of p-type emitters with a much higher sheet resistance, which in turn significantly enhances the front surface passivation by reducing Auger recombination in the heavily doped regions. As a result, laser-enhanced contact technology improves open-circuit voltage and overall cell efficiency, elevating the silicon solar cell performance. Meanwhile, the LASER exhibit

I don't know this physics yet

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excellent stability [4, 5], making it a suitable process to be incorporated in cell fabrication. Given its significance and relevance for fabricating high-efficiency solar cells, in this study we present a simulation-based analysis focused on quantifying the impact of a laser-enhanced contact approach on the performance of a typical n-type TOPCon cell, with a front boron-doped emitter and rear phosphorus-doped polysilicon on ultrathin silicon oxide interlayer (poly-Si/SiO<sub>x</sub>) structure.

Concurrently, as surface passivation improves, the quality of industrial Czochralski (CZ) grown Si wafers becomes increasingly critical for achieving high-efficiency solar cells. Although the level of transition metal impurities has been significantly reduced during the ingot growth [8–10] of modern CZ-Si and can be further reduced through gettering in some solar cell fabrication process [11–14], the bulk silicon materials may still pose a limit on the device efficiency. To assess the quality of industrial CZ-Si wafers and its impact on cell performance, we first employed injection dependent lifetime spectroscopy (IDLS) [15–18] to extract the electrical properties of the defects in such silicon wafers after gettering through polysilicon formation [12–14, 19], which represent the residual defects after cell processing since some impurities can be gettering during the solar cell fabrication process. We then applied these apparent residual bulk defect parameters to the simulation of cell performance.

In addition, the optimal silicon bulk resistivity for cell performance remains a topic of ongoing discussion, and is affected by various factors, such as silicon material quality, processing conditions, and device architecture. A deeper understanding of these interactions is crucial for refining solar cell design and maximizing overall efficiency. This study presents a simulation-based analysis of TOPCon devices with either SE or LASER on the front side and assessed the effects of bulk resistivity and bulk defect density on device performance.

It is also important to note that solar cell efficiency and the power ratings of photovoltaic (PV) modules are typically reported under standard test conditions (STC), which assume 1-sun illumination (1000 W/m<sup>2</sup>) at 25°C. However, in real-world applications, PV modules rarely experience continuous 1-sun illumination and may operate under various climate conditions, where the annual energy yield of PV systems may be strongly influenced by the low-illumination characteristics of the cells [20, 21]. To address this, we also compared the low-illumination response of the two types of devices (conventional SE and laser-enhanced contact),

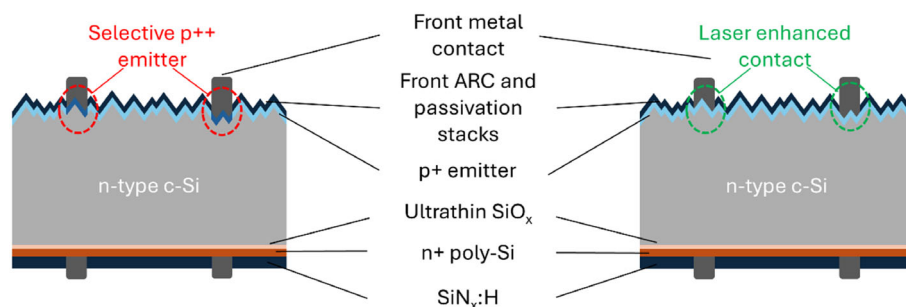
exploring their performance across a range of bulk resistivities. In addition, monthly and annual energy output of the two types of devices for representative locations in Australia were simulated using historical irradiance data with a 5-min resolution, providing a comprehensive evaluation of the impact of local solar resource variations on energy yield.

Overall, this work is to evaluate the performance of silicon solar cells with laser-enhanced contacts through comprehensive device simulation analysis, shedding light on the potential benefits and limitations of this advanced technology.

## 2 | Methodology

The samples used in this study for wafer quality assessment were industrially sourced phosphorus-doped n-type and gallium-doped p-type silicon wafers with different resistivities: n-type 0.5 Ωcm, n-type 0.8 Ωcm, n-type 1.2 Ωcm, p-type 1.2 Ωcm, p-type 1.6 Ωcm, and p-type 2.0 Ωcm. After standard RCA cleaning, the samples were symmetrically passivated by n-type phosphorus-doped poly-Si/SiO<sub>x</sub> structures and capped with aluminum oxide and silicon nitride on both sides, resulting in a surface recombination current density ( $J_0$ ) of around 1.5 fA/cm [2] per side. The injection-dependent lifetime measurements were performed using a Sinton WCT-120 instrument under the photoconductance decay method [22]. The intrinsic recombination was calculated with the recent radiative [23] and Auger recombination [24] models. After accounting for radiative, Auger and surface recombination components, IDLS analysis [15] was applied to extract the SRH parameters of the apparent bulk defect, assuming that it is a recombination active bulk defect with only one single energy level.

The two cell structures simulated in this study are high-efficiency n-type poly-Si/SiO<sub>x</sub> based silicon solar cells with different front p+ emitters as shown in Figure 1: one is a selective emitter design with local heavily doped regions under front metal fingers (referred to as “SE”), and the other features a full area lightly doped p+ emitter with laser enhance contact technology (referred to as “LASER”). The simulations were performed with Quokka 3 [25], with the simulation inputs listed in Table S1. The electrical inputs were estimated from the literature [26–28], with details noted in Table S1. The optical model of the cells is based on a lumped front surface transmission extracted from SunSolve Power. To generalize the simulation-based comparison between SE and LASER technologies, device simulations were performed



**FIGURE 1** | Schematic cross-section of the n-type poly-Si/SiO<sub>x</sub> based solar cells, with a structure of either SE (left) or laser enhanced contact (right) on the front side.

on the same cell geometry (e.g., silicon substrate thickness, finger width, finger pitch, etc.), but with varying bulk resistivities and defect densities. Typical SE and LASER cells are known to have slightly different processing conditions for the boron diffusion and firing steps, which may, in principle, lead to slightly different levels of ring defects and hydrogen-related defects. However, such defects are not prevalently observed in completed solar cells, and therefore their impact on the final bulk properties is considered small. We therefore assume the same bulk properties for the SE and LASER cells in some of the subsequent simulation scenarios.

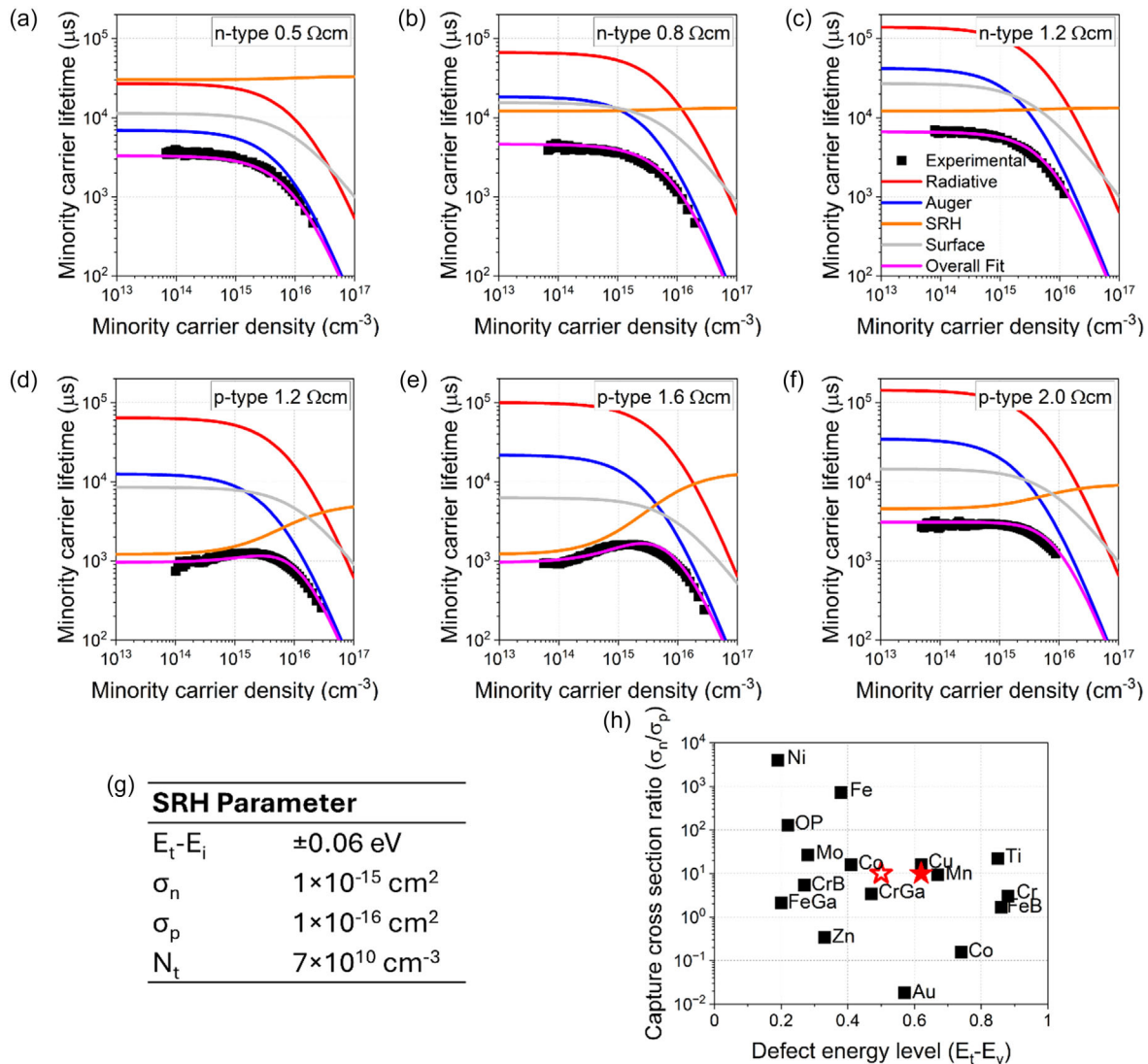
The energy yield simulation of device output was performed by SunSolve Yield using the solar irradiance data with a temporal resolution of 5 min [29], with the assumption that a 144-half-cell module performs under sunlight with single-axis tracking. Electrical input parameters were derived from Quokka 3 generated IV curves for varying light intensities and bulk resistivities, with the inclusion of the apparent bulk defects (details below).

The impact of module temperature was considered in the module energy yield simulation, assuming a typical temperature coefficient of  $-0.29\%/^{\circ}\text{C}$  for both types of devices. The monthly and annual energy yield was examined at three representative locations covering a range of annual irradiance distributions: Alice Springs with consistent and high irradiance, Wagga Wagga with varied and moderate irradiance, and Melbourne with the worst and largely varying solar irradiance.

### 3 | Results and Discussion

#### 3.1 | Experimental Assessment of the Wafer Quality

Figure 2a–f presents the measured and simulated injection-dependent lifetime curves of the six sets of samples. The radiative, Auger, and surface recombination components of the



**FIGURE 2** | (a–f) Experimental and simulated injection-dependent lifetime curves for six sets of samples: n-type 0.5  $\Omega\text{cm}$ , n-type 0.8  $\Omega\text{cm}$ , n-type 1.2  $\Omega\text{cm}$ , p-type 1.2  $\Omega\text{cm}$ , p-type 1.6  $\Omega\text{cm}$ , and p-type 2.0  $\Omega\text{cm}$ . The simulated lifetimes include radiative [23], Auger [24], surface, and SRH components, with the SRH component being the only fitting component. (g) The generalized SRH parameters extracted from the IDLS for all six sets of the samples. (h) A plot of capture cross-section ratio ( $k = \sigma_n / \sigma_p$ ) versus defect energy level for the extracted SRH parameters (red star: solid and open, as the specific half of the bandgap cannot be determined) and the reported SRH parameters of common defects (black square) in silicon wafers [15, 30–39].

injection-dependent lifetimes were calculated based on known models and measured input parameters (detailed above), while the SRH lifetime was the only fitting component. The carrier lifetimes for all samples are shown to be limited by SRH recombination at injection levels below the  $10^{15}$ – $5 \times 10^{15} \text{ cm}^{-3}$  range, except the n-type  $0.5 \Omega\text{cm}$  samples where Auger recombination dominates the effective carrier lifetime due to the heavy background doping in the silicon wafer bulk. At higher injection levels, Auger recombination starts to play a more dominant role, manifesting the impact of bulk resistivity.

The detailed SRH parameters for each sample are listed in Table S2 with the least square errors, and as shown in Figure 2a–f, the simulated lifetime curves show a good agreement with the experiment data. The extracted capture cross-section ratio ( $k = \sigma_n/\sigma_p$ ) was found to be greater than unity ( $k > 1$ ), suggesting that the lifetime curves of the p-type samples are more sensitive to the bulk defects and exhibit a stronger injection-dependence compared to n-type samples, as can be seen in Figure 2a–f.

The generalized SRH parameters that fit for the majority of samples with minor variation are listed in Figure 2g, which were further utilized for device simulation. As can be seen, the defect energy level was found to be  $0.06 \text{ eV}$  from the intrinsic Fermi level,  $E_t - E_i = \pm 0.06 \text{ eV}$ , since we are unable to determine which half of the bandgap from room-temperature IDLS. It is worth noting that the extracted SRH parameters are applicable to both n-type and p-type samples, indicating that the observed bulk defects exhibit similar densities and characteristics regardless of the background dopant type or doping level, therefore unlikely to be caused by dopant-related defects or complexes.

Figure 2h shows a comparison between the extracted SRH parameters and the reported parameters of common defects in silicon [15, 30–39] by a plot of capture cross-section ratio ( $k = \sigma_n/\sigma_p$ ) versus defect energy level with respect to the valence band edge ( $E_t - E_v$ ). The best-matched candidates are found to be copper precipitates and manganese. However, manganese is generally not considered to be a main impurity in silicon devices [40] and there is little information about its presence or concentration in the silicon materials for PVs. On the other hand, copper precipitates have been observed in the silicon materials for PVs (although reported in multicrystalline silicon) [32, 41–43], resulting in significant carrier recombination [32, 44]. Nevertheless, as the samples were passivated by n-type phosphorus doped poly-Si/SiO<sub>x</sub> structure, which provides excellent gettering effects for metal impurities [12, 13], most of the fast or moderately fast diffusing metal impurities are expected to be removed from the silicon wafer bulk, including copper precipitates as well [45]. Although the gettering of copper precipitates first requires a precipitate dissolution step, given copper's high solubility and diffusivity in silicon [40], the gettering process is unlikely to be limited by precipitate dissolution. It is therefore more likely that the extracted SRH parameters come from unknown defects or a combination of multiple defects, that are not sensitive to gettering.

Despite the difficulty in exclusively identifying the exact bulk defect, the extracted SRH parameters represent the properties of an apparent defect in the bulk of industrial CZ-Si wafers. Since the apparent defect parameters were extracted from the samples after gettering, it is highly relevant for cell simulation

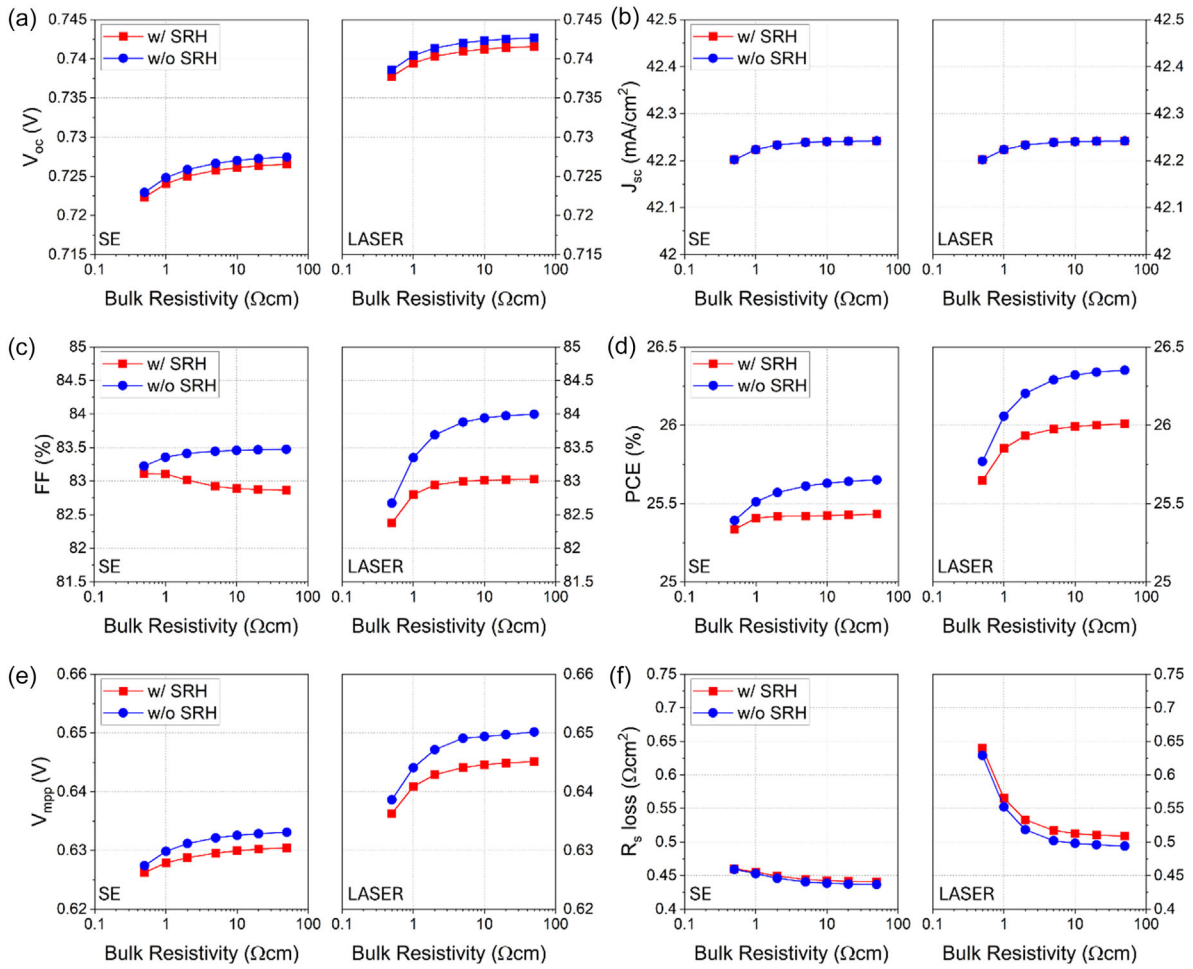
as it represents the remaining residual bulk defects that directly affect the final cell performance.

The extracted SRH parameters were then used for device simulations. Note that they were extracted from a relatively narrow bulk resistivity range of  $0.5$ – $1.2 \Omega\text{cm}$  for n-type ( $N_D = 4.1 \times 10^{15}$ – $1.2 \times 10^{16} \text{ cm}^{-3}$ ) and  $1.2$ – $2.0 \Omega\text{cm}$  for p-type ( $N_A = 7.2 \times 10^{15}$ – $1.2 \times 10^{16} \text{ cm}^{-3}$ ), but the subsequent simulation considers a wider range of bulk resistivities from  $0.3$  to  $50 \Omega\text{cm}$ , for an n-type silicon substrate. However, since the defect is unlikely to be related to dopants (discussed above), in this study the simulation analysis assumed that the defects remain unaffected by the variations in bulk resistivity.

### 3.2 | Impact of SRH Defect Density and Bulk Resistivity on Device Performance

Figure 3a–f compares the simulation results of TOPCon devices featuring either SE or LASER across a range of bulk resistivities, with or without the previously extracted SRH bulk defect parameters. As can be seen, the LASER devices exhibit generally superior performance to the SE devices, mainly resulting from an improved front surface passivation, as shown in the  $J_0$  parameters in Table S1. There are two main reasons contributing to the reduced  $J_0$  parameters: one is that the laser-enhanced contacts can significantly reduce Auger recombination on the front surface by lowering the required doping concentration, without sacrificing the contact resistance between metal and silicon [1, 2]; the other is that the laser process can also reduce recombination at the metal/silicon contact region, by applying a different metal paste of no/low aluminum fired at a lower temperature, forming localized microscale contacts under the metal fingers [5, 6]. Thus, as expected, the simulated LASER devices exhibit higher open-circuit voltage ( $V_{oc}$ ) and maximum-power-point voltage ( $V_{mpp}$ ), due to an improved minority carrier lifetime in the silicon wafer bulk, while the short-circuit current ( $J_{sc}$ ) remains similar to the SE devices as the same optical model was used in this study.

In addition, for both SE and LASER devices with or without SRH defects, the device performance ( $V_{oc}$ ,  $J_{sc}$ ,  $V_{mpp}$ ) improves with increasing bulk resistivity. This can be attributed to an improved minority carrier lifetime in high-resistivity silicon wafers, resulting from the alleviation of Auger recombination constraints through the reduction of the majority carrier concentration in the silicon bulk. Consequently, the minority carrier density increases, leading to an improved power conversion efficiency (PCE), as shown in Figure 3d. Meanwhile, as the bulk resistivity increases, the difference between the samples with and without SRH defects becomes more pronounced, as high-resistivity silicon wafers are more susceptible to SRH recombination. This was experimentally observed in Figure 2, where the effective carrier lifetime becomes increasingly limited by SRH defect lifetime for the n-type silicon wafers of a higher bulk resistivity. In addition, the difference in  $V_{mpp}$  between samples with and without SRH defects is more evident than in  $V_{oc}$ , as SRH defects have a greater influence at lower injection levels, as previously discussed. This highlights the critical demand for silicon wafers of higher bulk quality, particularly if the PV industry is



**FIGURE 3** | Simulated cell parameters for SE (left) and LASER (right) devices of various bulk resistivities with or without SRH defects: (a) open-circuit voltage, (b) short-circuit current, (c) fill factor, (d) PCE, (e)  $V_{mp}$ , and (f) series resistance loss.

considering moving toward high-resistivity silicon wafers for their benefit of higher intrinsic lifetimes.

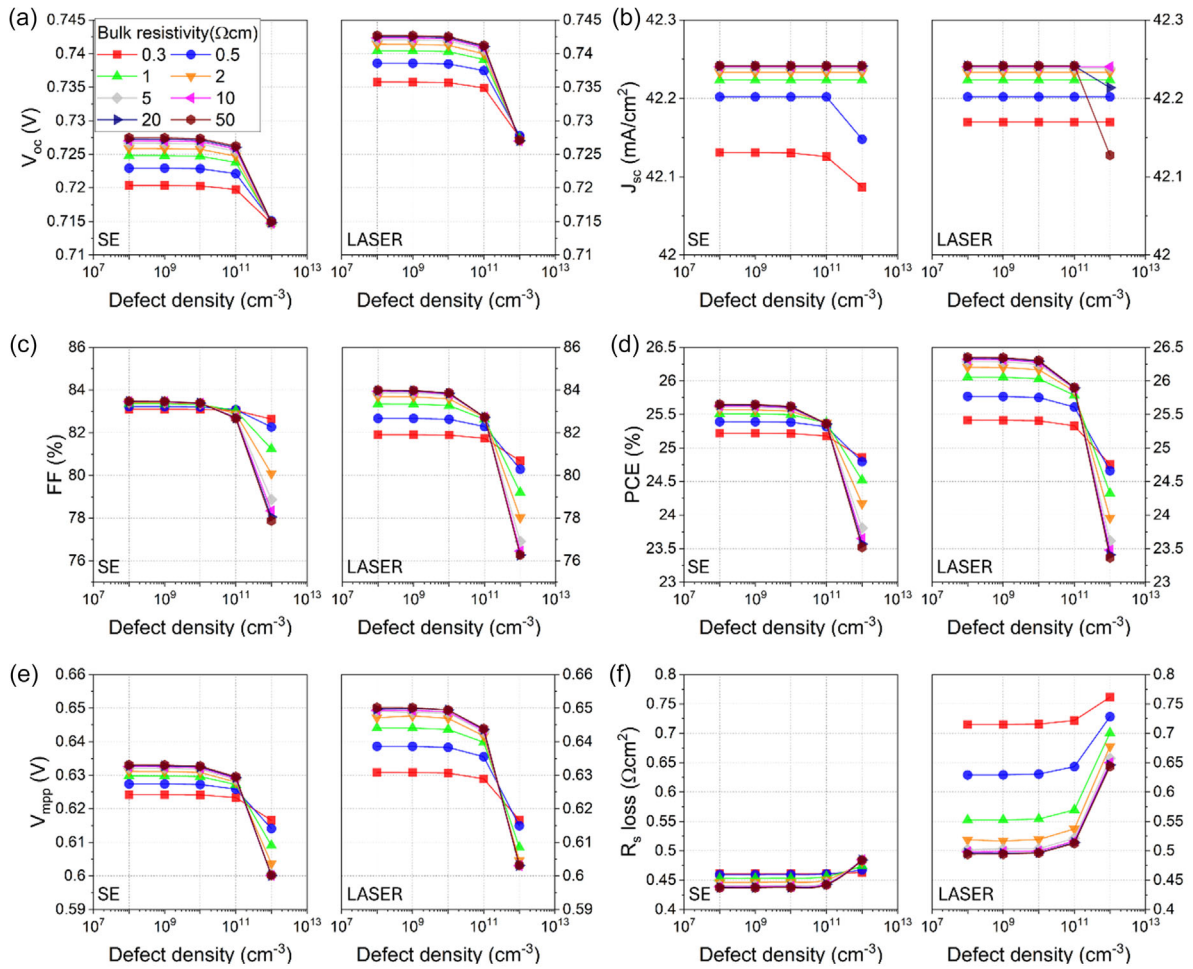
The simulation results for the fill factor (FF) reveal complex behaviors strongly influenced by SRH bulk defects, as shown in Figure 3c. In SE devices without SRH defects, FF increases with bulk resistivity due to enhanced carrier lifetime. However, in the presence of SRH defects, this trend reverses—FF decreases as bulk resistivity increases. In contrast, LASER devices show a consistent improvement in FF with increasing resistivity, regardless of SRH defect presence. Notably, a sharp FF drop is observed in LASER devices at resistivities of  $1.0 \Omega\text{cm}$  or lower, likely due to elevated series resistance losses (Figure 3f), which include both resistive (majority carrier) and diffusive (minority carrier) components [46].

The declining FF trend in SE devices with SRH defects aligns with previous reports [47, 48], attributed to enhanced SRH recombination. However, this behavior is not observed in LASER devices with SRH defects, suggesting that architectural differences—such as the absence of SE, the higher sheet resistance of the full-area p+ emitter, and improved front surface passivation—significantly affect FF behavior. To further explore this, simulations with modified SE parameters were performed (Figure S1).

The results from Figure S1b indicate that pseudo-FF (pFF) also decreases with increasing resistivity in the presence of SRH defects, supporting the observed FF trend in Figure 3c. Notably, changes to emitter sheet resistance—not contact recombination—were found to reverse this FF trend by mitigating series resistance losses (Figure S1c). The higher sheet resistance of the p+ emitter amplifies resistive losses, leading to overall higher series resistance across the resistivity range. As bulk resistivity decreases, reduced carrier lifetimes lower the minority carrier density, increasing diffusive losses and total series resistance.

These findings suggest that the reversed FF trend in SE devices arises from the interplay between SRH recombination and carrier transport limitations, while the relatively steady FF improvement in LASER devices is primarily constrained by increased resistive losses at low bulk resistivities.

Here, the impact of SRH defect density on the cell performance was analyzed using the previously extracted apparent SRH parameters. Additional details on the simulated impact of defect density are provided in Figure 4. As illustrated, both SE and LASER devices benefit from high-resistivity silicon wafers when the SRH defect density is low. However, as the defect density increases to the range of  $10^{11}$ – $10^{12} \text{ cm}^{-3}$ , the device performance deteriorates significantly due to SRH recombination. Notably,



**FIGURE 4** | Simulated cell parameters for SE (left) and LASER (right) devices of varying bulk resistivity and SRH defect density: (a) open-circuit voltage, (b) short-circuit current, (c) FF, (d) PCE, (e)  $V_{mp}$ , and (f) series resistance loss.

under high SRH recombination, devices with higher bulk resistivities exhibit even lower efficiency compared to those with lower bulk resistivity. Furthermore, LASER devices are found to be more sensitive to SRH defects, resulting in even lower efficiencies than SE devices at higher SRH defect densities.

This observation underscores the importance of higher-quality wafers for high-efficiency silicon solar cells, particularly as the PV industry shifts toward high-resistivity silicon wafers. While high-resistivity wafers can improve minority carrier lifetime and PCE by mitigating Auger recombination, these benefits depend critically on maintaining low levels of SRH defects. In the cases where SRH recombination is significant, low-resistivity wafers may still provide advantages.

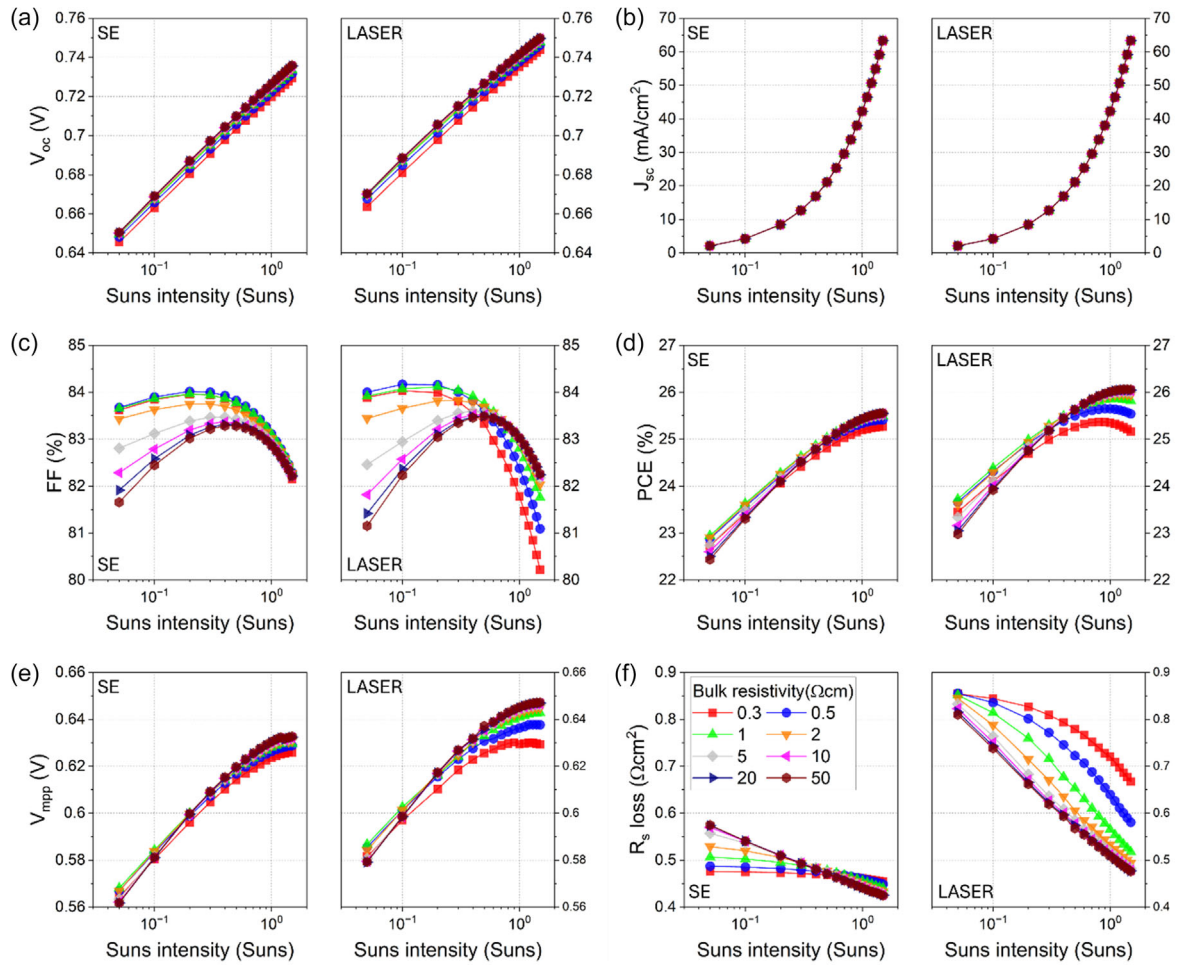
### 3.3 | Low-Light Illumination Response of the Device Performance

Figure 5a–f compares the simulation results for SE and LASER devices with varying bulk resistivities under different illumination intensities in the presence of SRH defects (parameter details listed in Figure 2g). As expected, the SE devices consistently exhibit inferior performance compared to LASER devices. With decreasing illumination intensity, the  $V_{oc}$  values of both

types of devices decline, and the devices of a higher bulk resistivity demonstrate slightly higher  $V_{oc}$  values. Notably, the slope of  $V_{oc}$  versus illumination intensity remains identical for both SE and LASER devices across different bulk resistivities. The  $J_{sc}$  values of both SE and LASER devices exhibit a good linear relationship with illumination intensity, which can be clearly observed when Figure 5b is presented with both axes plotted on a logarithmic scale (not shown here).

The FF values exhibit a distinct trend, initially increasing and then decreasing significantly as illumination intensity decreases, and both types of devices display a noticeable hump in the FF trend. Devices with higher bulk resistivities experience a more pronounced drop in FF values under low illumination conditions. In contrast, devices with lower bulk resistivities exhibit better low-light illumination response and maintain relatively stable FF values under similar conditions. Notably, LASER devices exhibit more pronounced degradation under low illumination, with the FF values of high bulk resistivity devices dropping below those of their SE counterparts. Similarly, the  $V_{mp}$  response follows the distinct FF trend at low illumination intensity, as shown in Figure 5e.

As can be seen from Figure 5f, the series resistance loss was found to increase with decreasing illumination intensity. This



**FIGURE 5** | Simulated cell parameters for SE (left) and LASER (right) devices of varying bulk resistivity under various illumination intensities with SRH defects: (a) open-circuit voltage, (b) short-circuit current, (c) FF, (d) PCE, (e)  $V_{mpp}$ , and (f) series resistance loss.

can be attributed to the increased diffusive loss (related to minority carriers) at low illumination intensities for both device types [46, 49], and the devices with a higher bulk resistivity exhibit more severe degradation in the series resistance loss. This increased series resistance loss can explain the drop of FF under low illumination intensity.

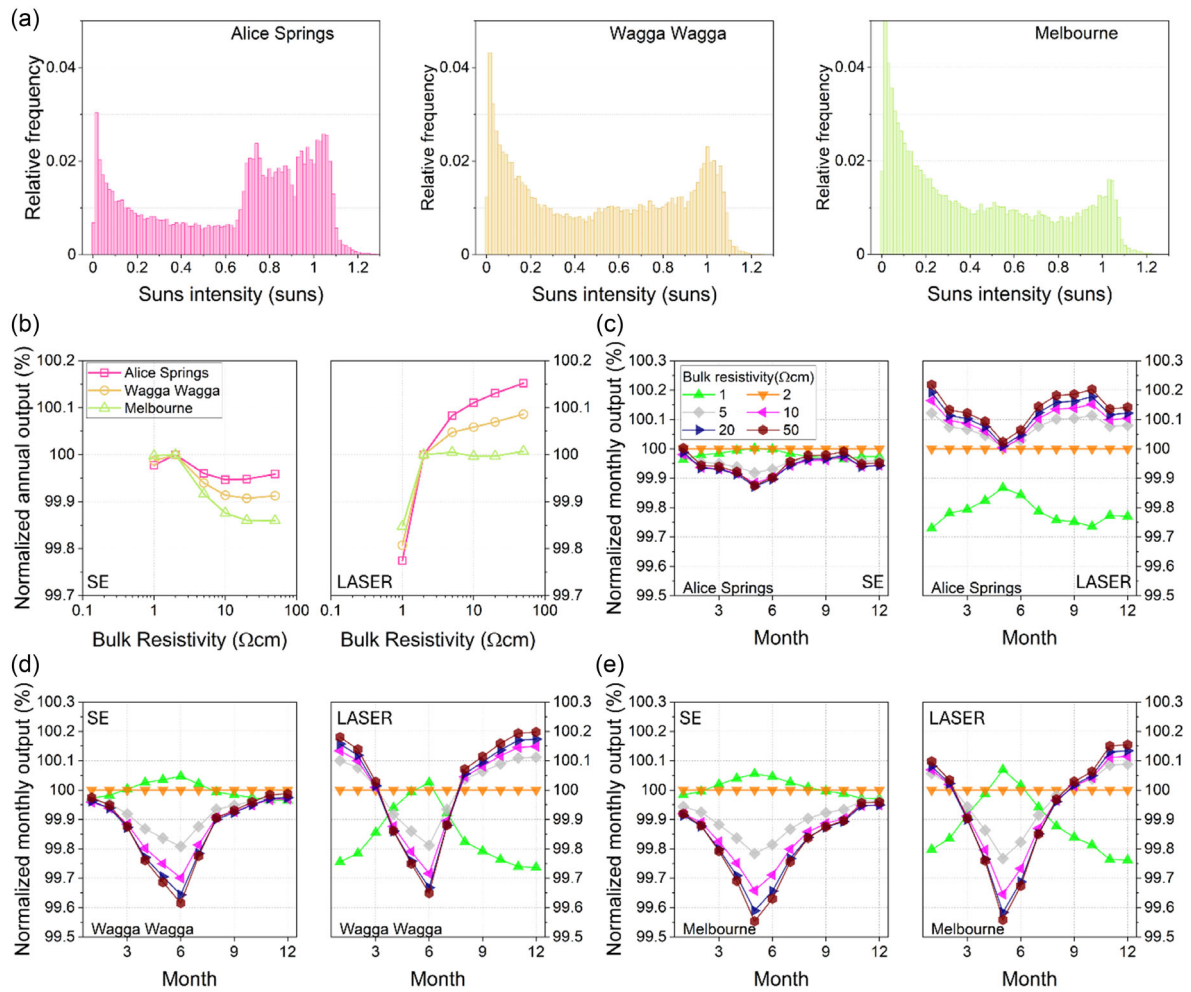
Another contributing factor is the presence of SRH defects, which intensify SRH recombination at low injection levels while Auger recombination dominates at high injection levels [16, 24], as shown in Figure 2. Notably, such a large FF drop is not observed in the simulated results without SRH defects (Figure S2). Under low illumination, while all devices experience increased series resistance loss, the loss is more pronounced in devices with SRH defects (Figure 5). Consequently, in the presence of SRH defects, devices with higher bulk resistivities experience a significantly greater increase in series resistance loss compared to those with lower resistivities, resulting in lower FF values under low-light illumination conditions.

It was previously deduced in Section 3.2 that a higher bulk resistivity leads to a higher device PCE under the standard one-sun condition, either with or without SRH defects. However, the results of low light illumination response show that, if the devices with SRH defects perform at a low

illumination intensity, the overall PCE of both types of devices with a higher bulk resistivity is inferior to the ones with a lower bulk resistivity (Figure 5d), while the devices without SRH defects benefit from high bulk resistivity across the whole light intensity range (Figure S2d). This distinguished difference highlights the complex nature of identifying a single optimum bulk resistivity for silicon solar cells, and the involvement of SRH defects in the degradation of device performance at various illumination intensities. Additionally, this raises questions regarding the overall performance of the devices under real-world sunlight conditions, which will be simulated and discussed below.

### 3.4 | Energy Output for Three Representative Locations in Australia

Module performance was further simulated for three representative locations in Australia (Alice Springs, Wagga Wagga, and Melbourne) to evaluate the energy output of both SE and LASER devices, with the inclusion of apparent SRH defect parameters and across different bulk resistivities. Figure 6a illustrates the relative frequency of the suns intensity at these locations. Alice Springs benefits from consistently abundant solar irradiance throughout the year, whereas Wagga Wagga



**FIGURE 6** | (a) Relative frequency of the suns intensity at three representative locations in Australia: Alice Springs, Wagga Wagga, and Melbourne. (b) Simulated normalized annual output for SE (left) and LASER (right) devices across different bulk resistivities, normalized to the results with a baseline resistivity of 2  $\Omega\text{cm}$ . (c–e) Simulated normalized monthly output for SE (left) and LASER (right) devices at (c) Alice Springs, (d) Wagga Wagga, and (e) Melbourne, normalized to the results with a baseline resistivity of 2  $\Omega\text{cm}$ .

experiences moderate irradiance, and Melbourne endures the poorest conditions out of the three locations, with low illumination prevailing.

Figure 6b displays the simulated annual outputs, normalized to a baseline resistivity of 2  $\Omega\text{cm}$ . SE devices achieve the peak performance at 2  $\Omega\text{cm}$ , with their annual output declining as resistivity increases, particularly under suboptimal solar conditions. In contrast, LASER devices benefit from a higher bulk resistivity, although the performance gains diminish in locations with poorer solar conditions. This trend aligns well with the device simulation results shown in Figure 3d, representing the device performance under the standard one-sun condition. Notably, the variations in annual output due to bulk resistivity remain within 0.2%, which is slightly lower than the relative PCE improvements observed in Figure 3d. For additional insights, Figure S3 provides the absolute monthly energy yields for devices with bulk resistivities of 2 and 50  $\Omega\text{cm}$ , illustrating seasonal fluctuations. The reduced energy yield improvement for high-resistivity devices is primarily attributed to low-light conditions, particularly during the winter months (June to August in Australia).

It is important to note that the simulation results in Figure 6 and S3 also account for the impact of module temperature on device performance, which is known to significantly impact device performance [50, 51]. However, the temperature correction has only a minor impact on the performance trends at a given location, as the same temperature coefficient was applied to both types of devices. This approach highlights the dominant role of irradiance variations in this study. For additional context, simulation results without temperature correction are provided in Figure S4 for reference.

Figure 6c–e presents the normalized monthly energy output for both SE and LASER devices across varying bulk resistivities, referenced to the 2  $\Omega\text{cm}$  baseline. In Alice Springs (Figure 6c), where solar irradiance is consistently high, the impact of bulk resistivity on SE device performance is minimal. In fact, a higher resistivity slightly reduces energy output compared to the baseline, aligning with the findings in Figure 3d (simulation based on the standard one-sun condition), which identified 2  $\Omega\text{cm}$  as the optimal bulk resistivity for SE devices with SRH defects. In contrast, LASER devices in Alice Springs demonstrate a steady increase in monthly output with rising bulk resistivity, following

the same trend observed in Figure 3d. This suggests that LASER devices can leverage the benefits of higher resistivity more effectively under strong solar conditions.

In contrast, devices operating in Wagga Wagga and Melbourne exhibit significant fluctuations in monthly output due to seasonal variations in solar irradiance between summer and winter. As depicted in Figure 6d–e, SE devices with higher bulk resistivities experience reduced monthly output, particularly during the winter months (June to August in Australia), when solar irradiance is at its lowest. While SE devices with a lower resistivity of 1  $\Omega\text{cm}$  show relatively better performance in winter, their output remains below the 2  $\Omega\text{cm}$  baseline even during summer. As a result, the annual output of SE devices in Wagga Wagga and Melbourne confirms an optimal bulk resistivity of 2  $\Omega\text{cm}$ , as depicted in Figure 6b.

Conversely, LASER devices in Wagga Wagga and Melbourne exhibit enhanced monthly outputs with increasing bulk resistivity during the Australian summer. However, during the winter, a pronounced decline in monthly output is observed, with performance falling below the 2  $\Omega\text{cm}$  baseline. This behavior is likely attributable to reduced efficiency under low-light conditions [20, 21], as illustrated in Figure 5d, which amplifies seasonal variations in energy output. Although this presents challenges for systems requiring a stable output profile, the overall annual performance of LASER devices still benefits from higher bulk resistivity—primarily due to improved PCE during summer months. However, the gains from higher bulk resistivities for LASER devices may be fully reversed if the solar conditions are even worse, as demonstrated in Figure S5 (a location with worse solar irradiance than Melbourne), where an optimal bulk resistivity was found to be 1  $\Omega\text{cm}$  for SE devices and 2  $\Omega\text{cm}$  for LASER devices, suggesting lower energy yields for both SE and LASER devices of high bulk resistivities.

Moreover, as shown in Figure S3c–d, the monthly energy yield comparison indicates that LASER devices achieve an improvement of over 2% relative to SE devices throughout the year, with even greater gains at a higher bulk resistivity of 50  $\Omega\text{cm}$ . This finding aligns with the trends presented in Figure 3d, underscoring the advantages of employing laser-enhanced metal contacts in n-type TOPCon solar cells.

## 4 | Conclusion

In this study, we first experimentally evaluated the quality of industrially sourced CZ-Si wafers and extracted the SRH parameters associated with the apparent bulk defects after gettering, which represent the residual defects present in the final device. Subsequently, we conducted a comprehensive simulation analysis based on the extracted SRH defect parameters to compare the performance of n-type TOPCon devices (boron doped p+ emitter on the front side and phosphorus doped poly-Si/SiO<sub>x</sub> passivating contacts on the rear side) either with conventional SE or incorporating laser-enhanced contacts on the front. The analysis examined device performance with and without SRH defects across a range of bulk resistivities (0.3–50  $\Omega\text{cm}$ ).

The results highlight the potential advantages of high-resistivity wafers in achieving high-efficiency silicon solar cells under STC, particularly when bulk defect densities are low. However, in the presence of significant bulk defect densities, high-resistivity wafers may lead to even lower efficiencies, especially in laser-enhanced contact devices, which are more susceptible to SRH recombination. This underscores the critical need for high-quality wafers with minimal SRH defect densities as the industry considers moving toward high-resistivity materials. Additionally, our energy yield analysis under low illumination conditions suggests that high bulk resistivity does not always confer a performance advantage, as its benefits are diminished in regions with poor or highly variable solar resources. Notably, while LASER devices demonstrated higher annual energy output compared to SE devices, they also exhibited larger seasonal variations, which may pose challenges for maintaining stable energy yields year-round. These findings provide valuable insights into the practical operation of TOPCon solar cells with laser-enhanced contacts, highlighting the importance of optimizing bulk resistivity and defect control to fully realize the efficiency and energy yield benefits across diverse environmental conditions.

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## Conflicts of Interest

The authors declare no conflicts of interest.

## Data Availability Statement

The data that support the findings of this study are available from the corresponding author upon reasonable request.

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## Supporting Information

Additional supporting information can be found online in the Supporting Information Section. **Supporting Information Figure S1:** Simulated (a,d) fill factor, (b,e) pseudo fill factor and (c,f) Rs loss for selective emitter devices of different bulk resistivities (a,b,c) with and (d,e,f) without SRH defects. Baseline: fill factor of baseline condition (i.e., the same as Figure 3); Front Rsheet: adjusting front full-area p+ emitter to a sheet resistance of 500  $\Omega/\text{sq}$  (200  $\Omega/\text{sq}$  in the baseline condition); SE Rsheet: adjusting front selective emitter with sheet resistance of 200  $\Omega/\text{sq}$  (80  $\Omega/\text{sq}$  in the baseline condition); Front+SE Rsheet: adjusting both front full-area p+ emitter and selective emitter both to a sheet resistance of 500  $\Omega/\text{sq}$ ; J0,metal: adjusting front metal/silicon recombination J0, metal to 75  $\text{fA}/\text{cm}^2$  (250  $\text{fA}/\text{cm}^2$  in the baseline condition). **Supporting Information Figure S2:** Simulated cell parameters for selective emitter (left) and laser-enhanced contacts (right) devices of different bulk resistivities under various illumination intensities without SRH defects: (a) open-circuit voltage, (b) short-circuit current, (c) fill factor, (d) power conversion efficiency, (e) maximum-power-point voltage, and (f) series resistance loss. **Supporting Information Figure S3:** (a,b) Simulated absolute monthly energy yield for selective emitter (left) and laser-enhanced contacts (right) devices of (a) 2  $\Omega\text{cm}$  and (b) 50  $\Omega\text{cm}$  at three representative locations in Australia with temperature correction. (c,d) Improvement in the simulated monthly energy output with temperature correction for the laser-enhanced contacts devices over selective emitter devices, with a bulk resistivity of (c) 2  $\Omega\text{cm}$  and (d) 50  $\Omega\text{cm}$ . **Supporting Information Figure S4:** (a,b) Simulated absolute monthly energy yield for selective emitter (left) and laser-enhanced contacts (right) devices of (a) 2  $\Omega\text{cm}$  and (b) 50  $\Omega\text{cm}$  at three representative locations in Australia without temperature correction. (c,d) Improvement in the simulated monthly energy output without temperature correction for the laser-enhanced contacts devices over selective emitter devices, with a bulk resistivity of (c) 2  $\Omega\text{cm}$  and (d) 50  $\Omega\text{cm}$ . **Supporting Information Figure S5:** Simulated normalized annual output for selective emitter

(left) and laser-enhanced contacts (right) devices at Hamelin, across different bulk resistivities, normalized to the results with a baseline resistivity of 2  $\Omega\text{cm}$ . **Supporting Information Table S1:** Input parameters for the Quokka device simulations on n-type silicon solar cells with either selective emitters or laser-enhanced contacts on the front side and full-area phosphorus-doped n-type polysilicon passivating contact on the rear. **Supporting Information Table S2:** The extracted SRH parameters from the injection-dependent lifetime spectroscopy for each set of samples.